PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Measurement and prognosis of frail patients undergoing transcatheter aortic valve implantation: a systematic review and meta-analysis
AUTHORS	Li, Zhe; Dawson, Emily; Moodie, Jessica; Martin, Janet; Bagur, Rodrigo; Cheng, Davy; Kiaii, Bob; Hashi, Adam; Bi, Ran; Yeschin, Michelle; John-Baptiste, Ava

VERSION 1 – REVIEW

REVIEWER	Hiroshi Furukawa Tokyo Women's Medical University Medical Center East, Japan
REVIEW RETURNED	02-Jul-2020

GENERAL COMMENTS	**Reviewer comments** This manuscript is well written and summarized the clinical role and implication of preoperative frailty in patients who underwent TAVI. This information will be of great importance in the current clinical setting. I congratulate the great and a lot of works to show the clinical evidence of frailty in TAVI patients. However, the manuscript seems to be complicated and not easy understand for readers. I have major and minor comments to correct your manuscript carefully for acceptance to publish in ***BMJ Open***. Please encourage considering and resubmitting following these comments. **Major comments:** 1. The form of manuscript is not unified and there are a lot of unnecessary space and mistyping. Please check carefully again. 2. This study is meta-analysis, but contents of this manuscript are too complicated to read. Please arrange the manuscript to the simple one. 3. This study included the unpublished data, but unpublished data is not used to quote to the meta-analysis. Please reconsider the study design again. **Minor comments:** 1. Page 7 line 89: frailt → frailty 2. Page 9 line 112: Two authors → show the initial of two authors 3. Page 9 line 118 ~ different font 4. Page 14 line 203 ~ 1, 2, 3 years survival, but show only two data?

REVIEWER	Carlos Rodriguez Pascual Complejo Hospitalario UNiversitario de Vigo Spain
REVIEW RETURNED	20-Jul-2020

GENERAL COMMENTS	I would loke to congratualate the authors for their work making this complex systematic review and metanalysis. I read the paper with interest. The main limitation of the analysis is beyend the author´s capacity, that is, the variability in frailty measures and toosl in literature and variability in frailty frequency in different publocations

REVIEWER	Samuel D. Searle Dalhousie University
REVIEW RETURNED	24-Jul-2020

GENERAL COMMENTS	Thank you for giving me the opportunity to review this manuscript which attempts to understand post TAVI outcomes of frail patients. The authors present a clear report of their methodological process and results. Highlighted - just like in much of frailty research - is the wide spectrum of frailty measures used and the difficulties that arise from different approaches to its operationalisation. Though not mentioned explicitly in the article, another assumption implicit in the research question, is that frailty is an all or nothing phenomenon. Because frailty and being robust exists on a continuous spectrum, having a firm cut point in defining those who are frail might introduce differences in study population demographics. Minor comments: line 168 appears to have only 3 frailty definitions, not 4. line 243: I happen to agree but wonder is 'confirms' is overly strong wording. I am also struct by the low/very low certainty (GRADE) of the studies. This study may provide additional useful evidence on prognosis of frail patients but cautiously so when taking into account the varied study populations and study biases.

REVIEWER	Prof Adrian Esterman University of South Australia Australia
REVIEW RETURNED	04-Aug-2020

GENERAL COMMENTS	I would like to see a better description of the statistical methods used, which don't completely match with the protocol paper. For example, the authors use a logistic transformation for dichotomous variables and reference paper 13, but that paper makes no mention of handling dichotomous variables. A better explanation would help readers. The Guyat et al reference is 15 in the paper, but 14 in the bibliography. A simple description is needed on how to interpret the funnel plot.

VERSION 1 – AUTHOR RESPONSE

Reviewer: #1

Major comments:
1. The form of manuscript is not unified and there are a lot of unnecessary space and mistyping. Please check carefully again.
Response: Thank you for raising this important issue. We have checked the form of manuscript and made corrections according to the BMJ Open format.

2. This study is meta-analysis, but contents of this manuscript are too complicated to read. Please arrange the manuscript to the simple one.
Response: Thank you for the comments. To make our manuscript easier to read, we added sub-headings in the Methods, Results and Discussion sections.
In the Methods section, we added 1) literature search and eligibility criteria, 2) risk of bias assessment, 3) data synthesis and meta-analysis, 4) subgroup analysis, and 5) GRADE assessment.
In the Results section, we added 1) characteristics of included studies, 2) measurement of frailty in patients undergoing TAVI, 3) Prognosis of frail TAVI recipients, and 4) GRADE assessment.
In the Discussion section, we added 1) principal findings, 2) comparison with other studies, 3) strengths and limitations, 4) implications, and 5) Conclusion.

3. This study included the unpublished data, but unpublished data is not used to quote to the meta-analysis. Please reconsider the study design again.
Response: Thank you for the comments. Given the value of grey literature, we believe including unpublished data would increase transparency and reduce publication bias. To be consistent, we applied the same eligibility criteria to citations from the grey literature. To clarify, we did include 2 unpublished data in the meta-analysis. Most unpublished data were excluded from meta-analyses because of overlaps in study populations.

Minor comments:
1. Page 7 line 89: frailt → frailty
Response: We have corrected this word.

2. Page 9 line 112: Two authors → show the initial of two authors
Response: We have showed the initials of the authors.

3. Page 9 line 118 ~ different font
Response: We have changed the font size to be consistent with the rest of the manuscript.

4. Page 14 line 203 ~ 1, 2, 3 years survival, but show only two data?

Response: We have reported survival data at 1, 2 and 3 years.

Reviewer: #2

Please leave your comments for the authors below
I would like to congratulate the authors for their work making this complex systematic review and metanalysis.
I read the paper with interest.
The main limitation of the analysis is beyond the author´s capacity, that is, the variability in frailty

measures and tools in literature and variability in frailty frequency in different publications

Response: Thank you for the comments. We agree that the variations in frailty measures lead to significant heterogeneity. This was an important finding in our review. We also acknowledged that some frailty measures were cited more frequently in the literature. We have summarized the frailty measures in the Results section.

Reviewer #3

Please leave your comments for the authors below
Thank you for giving me the opportunity to review this manuscript which attempts to understand post TAVI outcomes of frail patients. The authors present a clear report of their methodological process and results. Highlighted - just like in much of frailty research - is the wide spectrum of frailty measures used and the difficulties that arise from different approaches to its operationalisation.

Though not mentioned explicitly in the article, another assumption implicit in the research question, is that frailty is an all or nothing phenomenon. Because frailty and being robust exists on a continuous spectrum, having a firm cut point in defining those who are frail might introduce differences in study population demographics.
Response: Thank you for these comments. We agree that frailty might not be an all or nothing phenomenon. We also agree that dichotomizing frailty status might introduce bias. However, to be consistent with previous literature in which patients were dichotomized as frail and non-frail, our review did not include studies without a cut-off point. Most importantly, our review focuses on the prognosis of frail TAVI recipients. Given the nature of our research objective, we believe dichotomizing frailty status would be a more appropriate approach. To be transparent, we acknowledged this limitation in the Discussion as follows:
"…. Some studies defined an intermediate 'pre-frail' group, but we did not find sufficient data to synthesize outcomes for this important sub-group. Though less vulnerable than the frail group, pre-frail patients may be at higher risk than robust patients for experiencing adverse outcomes."

Minor comments:

line 168 appears to have only 3 frailty definitions, not 4.
Response: We have corrected this error and reference three frailty definitions.

line 243: I happen to agree but wonder is 'confirms' is overly strong wording.
Response: Thank you for the comment. We have rephased this sentence as below to make it more conservative.
"Thus, our study suggests that frailty is a multi-dimensional phenomenon that cannot be captured by a single construct."

I am also struct by the low/very low certainty (GRADE) of the studies.

This study may provide additional useful evidence on prognosis of frail patients but cautiously so when taking into account the varied study populations and study biases.
Response: Thank you for these comments. The very low certainty in the overall estimates was primarily caused by the variety of frailty definitions and the diversity of TAVI populations in the studies. After including more recent studies, we were able to provide additional evidence on quality of life in frail patients undergoing TAVI.

Reviewer #4

Please leave your comments for the authors below
I would like to see a better description of the statistical methods used, which don't completely match with the protocol paper. For example, the authors use a logistic transformation for dichotomous variables and reference paper 13, but that paper makes no mention of handling dichotomous variables. A better explanation would help readers. The Guyat et al reference is 15 in the paper, but 14 in the bibliography. A simple description is needed on how to interpret the funnel plot.
Response: Thank you for the comments. We have clarified the statistical methods as follows:
"…For adverse clinical outcomes, we pooled proportions using the inverse-variance weighted DerSimonian and Laird model and incorporated the Freeman-Tukey double arcsine transformation.13,14 A funnel plot was used to plot the effect estimates from individual studies against the standard error of each study. In the absence of bias and heterogeneity, the funnel plot will be symmetrical.17 For the length of hospitalization, we pooled the values, estimating the mean and standard deviation using the random effects model for continuous variables.15 For studies presenting Kaplan-Meier curves with time to death, we collected the information on numbers at risk and total number of events, and then created a single pooled Kaplan-Meier curve. We pooled time to death data from individual studies to obtain an overall estimate of survival, based on an algorithm developed by Guyot et al.16 All analyses were conducted using R software (version 3.5.0). A two-sided p-value of 0.05 or less was considered statistically significant."

VERSION 2 – REVIEW

REVIEWER	Prof Adrian Esterman University of South Australia Australia
REVIEW RETURNED	29-Nov-2020

GENERAL COMMENTS	Thank you for making the recommend changes.